

Model-based Meta Analysis in QSP, as an alternative method for producing virtual populations

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What we want, and what gets built

In QSP, when we are producing virtual populations, implicitly what we want is the distribution F of the model's parameters across real patients:

$$\theta_i \sim F.$$

, where each θ_i represents a virtual patient.

Everything we would want to report is computed from F . What fraction respond, how wide the response is, where the tail sits.

Classically, a **virtual population** is not that. It is a fixed set of candidate patients (that is, draws θ_i) with weights on them, chosen so that summaries computed from the weighted set match the summaries papers printed. So we are still estimating F in a sense, but heuristically.

Why that is a stand-in

$$\mathcal{P} = \{ \text{weightings } p : T(p) = \hat{T} \}$$

- About 50 published numbers against about 10,000 candidate patients, so $\mathcal{P}$ is enormous.
- The tie is broken towards the distribution the cloud was drawn from, which was made wide on purpose so the cloud would cover everything. Silent directions therefore come back that wide, which is wider than anyone believes.
- One point of $\mathcal{P}$ comes back, with no way to tell which directions the data pinned and which that starting distribution did.

Regularizing (falling back to a default) is not bad - we do it too. The objection is *which* prior, and what comes back.

What this model does instead

Put F in a family, and put a posterior on the family:

$$\theta_i \mid \phi \sim F(\cdot \mid \phi), \quad p(\phi \mid \mathcal{D}).$$

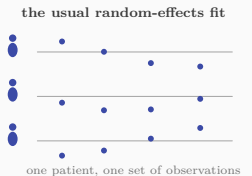
- The unknown is ϕ , a few hundred numbers, rather than a weighting on ten thousand patients.
- Each posterior draw $\phi^{(t)}$ is an entire population $F(\cdot \mid \phi^{(t)})$.
- The trade we make is that F has to lie in the family, and we pick the family by hand.

The work is all in $p(\phi \mid \mathcal{D})$, because of what $\mathcal{D}$ is.

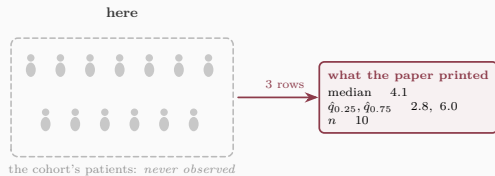
The problem: what $\mathcal{D}$ is

$$\hat{T} = T(\hat{F})$$

- $\hat{F}$ the $n_c \approx 10$ patients a paper had
- T the summary it computed, such as a median or quartile
- $\hat{T}$ the number it printed. We have about 54 of them

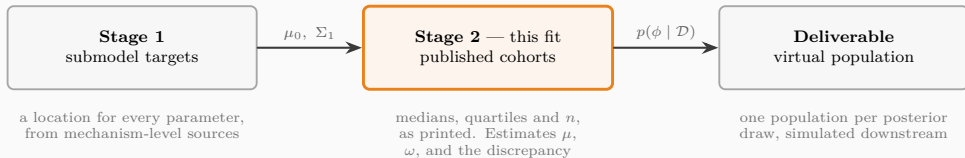


The likelihood factors over patients, and each patient's own scatter is what identifies the population spread.



Nothing factors. A row is a functional of the whole cohort, computed by someone else, under their own eligibility rule and their own assay.

Where this fit sits



- **Stage 1** already fixed roughly where each of the few hundred parameters sits. It says nothing about how much patients differ from one another, because its sources are not populations (typically in vitro experiments).
- This fit adds that spread, and lets the published cohorts move the center as well.

What one paper gives us

A paper biopsies ten patients on day 21 and reports the density of CD8^+ T cells in viable tumour, as a median and two quartiles.

median	4.1
lower quartile	2.8
upper quartile	6.0
patients	10

Counted the way this model counts them, that is **three rows** of data rather than one. The whole dataset is about 54 rows, scraped from a wide variety of papers with MAPLE

This demonstrates the dimensionality differences: Three numbers off ten people, and from roughly fifty such triples we are trying to recover a few hundred parameters and how much each of them varies between patients.

Some vocabulary

- readout** the quantity a paper reports. Here, the density of killer T cells.
- cohort** the group of patients it was reported on, with its own size and its own rule about who was allowed in.
- row** one readout, from one cohort, summarized one way. The median above is a row, and each quartile is another.
- scenario** the situation the simulator is run in so that it matches the cohort: untreated, say, or three weeks into treatment.
- block** a set of cohorts that have to be handled together, because they were measured on overlapping patients.

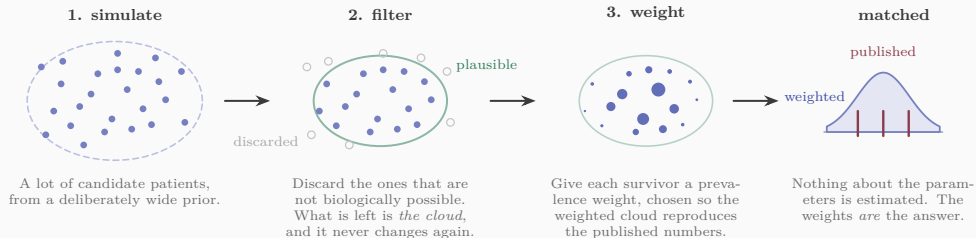
How virtual populations get built: a comparison of methods

approach	what it needs	what it returns
generate and select plausible patients free: which candidates are kept	a range per observable	 a set of patients, none more common than another
reweight a fixed cloud free: prevalence weights on the cloud	observed marginals	 one weighting, and no band around it
fit patients individually free: one θ per real patient	individual patient data	 one posterior <i>per patient</i>
this model free: (μ, ω) and the discrepancy	cohort summaries, with n	 a <i>distribution over populations</i> , one per posterior draw

Cheng et al. 2017 (QSP Toolbox); Rieger et al. 2018; Allen, Rieger & Musante 2016; Paul et al. 2024. Row two is `vpop/weighting.py`.

Deeper into the second method (Allen/Rieger)

The first and second rows are the standard methods in this field. In particular, the second works like this:



The parameter values are settled at step two and never touched again. All the data decide is how common each of those already-chosen patients is.

Diagnostics from this method

$$\text{ESS} = \left(\sum_m p_m^2 \right)^{-1}$$

In my experience fitting this on the PDAC model, this collapses. A handful of the 10,000 candidates end up carrying nearly all the mass, and there are two reasons for it.

- **Dimension.** One weight vector has to satisfy every readout at once, and the feasible fraction of the cloud shrinks with the number of readouts even when the model is right.
- **Miscalibration.** The model cannot reproduce the readouts jointly, so the only survivors are cloud members out in the tail.

These call for opposite remedies: a bigger cloud or a better model.

What a likelihood needs

$$\hat{T} \sim \mathcal{N}(\tau(\phi), V)$$

$\tau(\phi)$ is the model's prediction of the printed number, and V sets the (co)variation in τ .

Four nuances/challenges about this likelihood, in the order the rest of the talk takes them:

1. τ , the model's prediction, is not the population quantity. That median came off ten patients, so we predict the average of a ten-patient median - so somehow we have to link from the sample quantity to the population quantity.
2. Nothing hands us V . Papers report spreads, not the uncertainty in their own summaries, so the model has to predict V from its own patients.
3. Not all of the bias $\hat{T} - \tau$ is biology. Some is the assay and some the simulator, and both need somewhere to go other than μ and ω - otherwise those parameters end up absorbing technical data issues.
4. The rows are not independent. One paper reports several numbers off the same patients, so V is block diagonal - capturing that numbers reported from the same study (typically) come from the same patients.

First: ten patients do not have the population's median

$$\tau = T(\tilde{F}) \quad \text{versus} \quad \tau = \mathbb{E}[T(\tilde{F}^*)], \quad \tilde{F}^* \text{ an } n_c\text{-sample from } \tilde{F}.$$

The model has to do what the paper did: take n_c of its own patients, compute their median, and average that over draws. This captures the generative process at work here - a patient sample comes from a larger population, which is what we are trying to estimate

Sample quartiles are biased inward, at both ends, so a printed pair is narrower than the population it came from. If we predict population quartiles against it, the model looks too wide.

Second: a spread is not an error bar

$$\hat{T} \sim \mathcal{N}(\tau(\phi), V), \quad V = ?$$

Papers report spreads readily enough. An IQR says how much patients vary; it does not say how far the median would move with a different ten. So the model predicts V by resampling its own cohort:

$$V^{\text{boot}} = \text{Cov}_* [T(\tilde{F}^*)].$$

But $\tilde{F}$ depends on ϕ . So V is built once at a preliminary ϕ_0 and then frozen, and the fit cannot chase its own error bars.

What remains is a dependence on ϕ_0 . It is allowable because V only weights the residuals: any fixed choice leaves the solution in the right place, so ϕ_0 costs precision and not correctness.

Third: some of the disagreement is not biology

$$\tau\left(\underbrace{\mu, \omega}_{\text{what we want}}\right) \quad \text{versus} \quad \tau\left(\underbrace{\mu, \omega}_{\text{what we want}}, \underbrace{a, b, \beta}_{\text{assay, simulator}}\right)$$

Two labs with different equipment do not report the same number, and the simulator is wrong somewhere as well.

With no term for it, that disagreement goes into μ and ω . With one, a , b and β can absorb real biology just as easily as an artefact.

Fourth: the rows are not independent

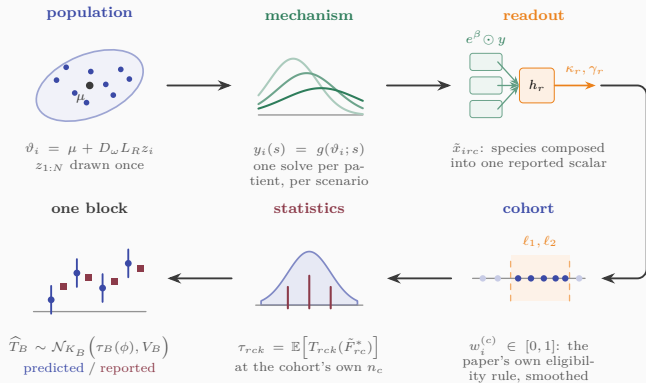
$$\prod_{k=1}^K \mathcal{N}(\hat{T}_k; \tau_k, V_{kk}) \quad \text{versus} \quad \prod_{B \in \mathcal{B}} \mathcal{N}_{K_B}(\hat{T}_B; \tau_B, V_B)$$

About 54 rows, and one paper supplies 20 of them off roughly ten patients per arm. The left product counts those ten people twenty times.

So the unit of independence is neither the row nor the cohort. It is the **block**: cohorts that have to be handled together, either because one row is computed from several of them or because they overlap in patients.

The generative model

The chain, end to end



Everything between the ends is deterministic given the fixed cloud $z_{1:N}$. The only stochastic step is the last one.

The whole model: parameters to a row

population $\vartheta_i = \mu + D_\omega L_R z_i$, $z_{1:N}$ drawn once

mechanism $y_i(s) = g(\vartheta_i; s)$

readout $x_{irc} = h_r(y_i(1), \dots, y_i(S))$

discrepancy $\tilde{x}_{irc} = \kappa_r(h_r(\mathbf{e}^\beta \odot y_i(s(c)))) - c_{rc}) + c_{rc} + \gamma_r$

with $\gamma_r = Z_r^\top \mathbf{a}$, $\log \kappa_r = Z_r^\top \mathbf{b}$

eligibility $\mathbf{w}_i^{(c)} = \sigma((\tilde{x}_{irc} - \ell_1)/h_w) \sigma((\ell_2 - \tilde{x}_{irc})/h_w)$

statistic $\tau_{rck}(\phi) = \mathbb{E}[T_{rck}(\tilde{F}_{rc}^*)]$, $\tilde{F}_{rc}^*$ an n_c -sample from $\tilde{F}_{rc}$

observation $\hat{T}_B \sim \mathcal{N}_{K_B}(\tau_B(\phi), V_B)$, independent across blocks

study effect $\eta_s \sim \mathcal{N}(0, \tau_\eta^2)$ on study s 's location rows, marginalised into V_B

error bars $V_B = V_B^{\text{boot}} + E_B + \tau_\eta^2 \sum_s \mathbf{1}_{B,s}^{\text{loc}} \mathbf{1}_{B,s}^{\text{loc}^\top}$, built at ϕ_0 and frozen

Here first, so the shape is visible. Each line gets its own slide after this.

The whole model: priors, posterior, deliverable

centre $\mu \sim \mathcal{N}_P(\mu_0, \Sigma_1)$, from stage 1

spread $\log \omega_j = \log \omega_{0j} + s + u_j$, $\sum_{j \notin \mathcal{M}} u_j = 0$, ω_0 elicited by role

correlation $R = \Lambda \Lambda^\top + \text{diag}(\psi^2)$, Λ elicited

assay $a \sim \mathcal{N}(0, \sigma_a^2 I)$, $b \sim \mathcal{N}(0, \sigma_b^2 I)$

species $\beta_q = \tau_\beta (\beta_q^{\text{raw}} - \bar{\beta}^{\text{raw}})$, τ_β fixed

conversions $\log R_m \sim \mathcal{N}(\log R_{0m}, \sigma_{Rm}^2)$

unknowns $\phi = (\mu, s, u^{\text{raw}}, \log \omega_{\mathcal{M}}, a, b, \beta_S^{\text{raw}}, \log R_{\mathcal{A}})$

posterior $p(\phi \mid \mathcal{D}) \propto p(\phi) \prod_{B \in \mathcal{B}} \mathcal{N}(\hat{T}_B; \tau_B(\phi), V_B)$

deliverable $p(\theta_{\text{new}} \mid \mathcal{D}) = \int \mathcal{N}_P(\log \theta_{\text{new}}; \mu, \Omega) p(\phi \mid \mathcal{D}) d\phi$

Above the gap, what we bring to the problem. Below it, what we solve for and what comes out.

Population

A patient is a parameter vector. The population is a lognormal over them:

$$\vartheta_i \mid \mu, \omega \sim \mathcal{N}_P(\mu, \Omega), \quad \Omega = D_\omega R D_\omega, \quad \vartheta = \log \theta,$$

with μ the centre and ω the spread of each parameter. ω sets the marginals; R sets everything joint.

R is not fitted: about 50 rows cannot inform $P(P-1)/2$ correlations. It is elicited, and deliberately not inherited from stage 1.

- A stage-1 posterior correlation is epistemic: the same data constrained those two jointly. Reading it here asserts something else and unevidenced, that a patient high in one is high in the other.
- So each parameter loads on a few latent patient axes, and R is what the shared loadings imply:

$$z_{ij} = \sum_k \Lambda_{jk} f_{ik} + \psi_j e_{ij}, \quad R = \Lambda \Lambda^\top + \text{diag}(\psi^2).$$

Two elicited inputs, one form

Nothing measures a between-patient spread, and nothing measures a between-patient correlation. Both are stated as claims about single parameters, in bins, each bin named for its **warrant** rather than its value.

Role: how wide, $\rightarrow \omega_0$

	ω_0	#
patient_trait	0.70	18
lumped	0.50	50
default	0.35	3
packing_limit	0.15	12
material_constant	0.10	4
structural	0.05	15

Loading: what it varies with, $\rightarrow \Lambda$

	$ \Lambda_{jk} $	explains
primary	0.8	64%
secondary	0.5	25%
weak	0.3	9%

No declared loading means uncorrelated, and that is the default. Communality capped at 0.81, so R is strictly positive definite and no nearest-correlation projection is ever called.

$O(P)$ claims rather than $P(P-1)/2$. The gaps are load-bearing too: $\tau_\omega = 0.085$ is a quarter of the narrowest, so the fitted pattern cannot move a parameter out of the role it was given.

We draw N standard normal vectors *once*, at the start, and reuse the same ones for every (μ, ω) the fit tries:

$$\vartheta_i = \mu + D_\omega L_R z_i, \quad R = L_R L_R^\top.$$

- Redrawing them each time would make the predicted rows jitter, and the fit would chase that jitter instead of the parameters.
- Reusing them also keeps patient i the same person across scenarios, which is what lets a readout compare a patient against themselves.
- With hundreds of parameters nothing gradient-free is in reach, so every step from ϕ to a predicted row needs a derivative. This is the first of three places that constraint decides a modelling choice. The others are enrolment and the quantile.

Mechanism and readout

Simulate patient i under scenario s , read the species at the measured time, and compose them into the scalar a study reports:

$$y_i(s) = g(\vartheta_i; s) \in \mathbb{R}_+^Q, \quad x_{irc} = h_r(y_i(1), \dots, y_i(S)).$$

- Most readouts use only $y_i(s(c))$. h_r takes the whole scenario set anyway, so that a fold change is a contrast within one patient.
- h_r carries the unit conversions, and is already fixed by the target definitions.
- In the fit, g is a trained emulator $\hat{g}$ rather than the solver itself. The error that introduces is priced in later.

Discrepancy

Two terms, separated by where they enter: one on the species, one on the finished row.

$$\tilde{x}_{irc} = \kappa_r \left(h_r \left(\underbrace{e^\beta \odot y_i(s(c))}_{\text{upstream of } h_r} \right) - c_{rc} \right) + c_{rc} + \gamma_r, \quad \gamma_r = Z_r^\top a, \quad \log \kappa_r = Z_r^\top b.$$

- $\beta \in \mathbb{R}^Q$ biases each *species* on the log scale, shared across readouts and scenarios.
- a and b are the location and scale halves of an affine map on the *readout*. Z_r holds assay modality and quantity type.
- $\beta = 0, a = 0, b = 0$ is a model biased in neither location nor scale on any observed species.

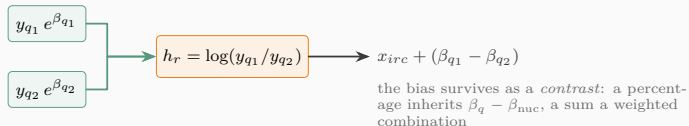
Why the two are separable at all

A free offset per row would fit everything and mean nothing. These two are earned by where they enter, because that decides how they propagate.

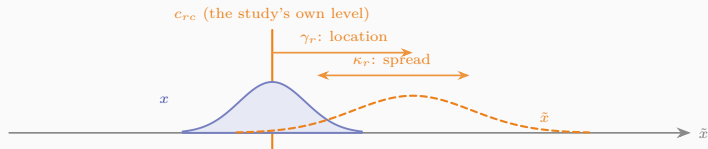
- β sits upstream of h_r , so the composition carries it: a ratio inherits $\beta_{q_1} - \beta_{q_2}$, a percentage inherits $\beta_q - \beta_{\text{nuc}}$, a sum inherits a weighted combination.
- γ_r sits downstream, so it cannot decompose. It shifts the finished number and does nothing else.
- One species biased at the source therefore leaves a coordinated pattern across every readout that touches it, in ratios the compositions fix in advance. A readout-level term cannot produce that pattern.

Where each one enters

upstream of h_r : a bias on the *species*



downstream of h_r : an affine map on the *row*, pivoting at c_{rc}



$\gamma_r = Z_r^\top a$ and $\log \kappa_r = Z_r^\top b$ read attributes of the readout with no mechanistic counterpart (assay modality; density, fraction or concentration). They act on the finished row, so they do not decompose through a ratio.

The pivot: what does κ_r stretch about?

Stretching needs a centre to stretch about. We use c_{rc} , the level readout r sits at in cohort c , computed once from the prior and never fitted.

- Stretch about zero instead and κ_r moves the readout as well as widening it, because a log density sits nowhere near zero. It would be doing γ_r 's job as well as its own.
- Stretching about the study's own level keeps them apart: κ_r is purely spread within the cohort, γ_r purely where the cohort sits.
- Every cohort gets its own c_{rc} , and that costs no parameters, because those levels are fixed before the fit starts.

Papers enrol patients on a criterion, so the model has to as well. A rule like “baseline density between ℓ_1 and ℓ_2 ” becomes a weight between 0 and 1 on each simulated patient, with a logistic σ standing in for the hard cut-off:

$$w_i^{(c)} = \sigma((\tilde{x}_{irc} - \ell_1)/h_w) \sigma((\ell_2 - \tilde{x}_{irc})/h_w).$$

- Soft because a hard cut-off has no derivative: one patient crossing the line would step the prediction. This is the second of the three.
- The rule is written in reported units, so it reads $\tilde{x}$, the value *after* the assay bias. Who is eligible therefore shifts as a , b and β shift. Enrolment is part of the fit, not fixed beforehand.

From a cloud to a row

A paper prints the median of nine patients. What do we predict?

Not the population median. The median of nine draws is a random variable, and the model has to predict its *mean*:

$$\tau_{rck}(\phi) = \mathbb{E} \left[T_{rck}(\tilde{F}_{rc}^*) \right], \quad \tilde{F}_{rc}^* = \text{an } n_c\text{-sample from } \tilde{F}_{rc}.$$

Same functional, same sample size, computed on the model's own predicted patients. That expectation is what τ is: the mean of the row, not the population quantity the row is trying to estimate.

Why the distinction is not academic

- Take a population with log-scale spread 1. Its lower quartile is -0.674 . Draw nine patients from it, and the lower quartile you compute averages -0.570 : pulled towards the middle, and the upper quartile is pulled in too.
- So a model predicting *population* quartiles can only meet a reported pair by shrinking its spread to **0.845**. That is a 15% error in the one number this whole fit exists to produce.
- The gap closes as the cohort gets bigger, not as the model's own cloud gets better, so nothing else in the fit can absorb it.

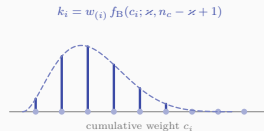
The population quantity is still what we deliver. It is read off the posterior predictive at the end, not off any row.

And the estimator has to be differentiable

$$\hat{q}(p) = \frac{\sum_i k_i x_{(i)}}{\sum_i k_i}, \quad k_i = w_{(i)} f_B(c_i; \varkappa, n_c - \varkappa + 1), \quad c_i = \frac{\sum_{l \leq i} w_{(l)} - \frac{1}{2} w_{(i)}}{\sum_l w_l}.$$



One patient carries the whole gradient. Two patients swap rank and it jumps.

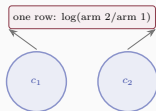


The rank of the $\varkappa$ -th of n_c draws is $\text{Beta}(\varkappa, n_c - \varkappa + 1)$ whatever the distribution, so centre $\varkappa/(n_c + 1)$ and width $\sqrt{p(1-p)/n_c}$ are set by n_c , not chosen.

The rank of the order statistic a cohort's rule points at is $\text{Beta}(\varkappa, n_c - \varkappa + 1)$ whatever the population is, so these weights are not a smoothing knob: they compute exactly the expectation above, and differentiability comes free. A width is out of the kernel's reach, though. An sd or an IQR is no order statistic, so it enters as its log, and that log is what makes κ_r an additive shift on the row.

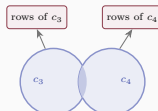
$$\hat{T}_B \sim \mathcal{N}_{K_B}(\tau_B(\phi), V_B), \quad \hat{T}_B \perp \hat{T}_{B'} \text{ for } B \neq B'.$$

joined by a shared row



Disjoint people. Each cohort is resampled independently; the correlation enters because one row reads both.

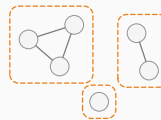
joined by shared people



shared patients

Resampled once from the block's patient set, each cohort's rows evaluated on its own members. Sharing need not be nesting: the block carries the union and each cohort keeps its n_c .

$\mathcal{B}$: connected components



Under both relations at once.
 $|B| = 1$ recovers a single cohort.

The Gaussian is a working model for a statistic of ten patients rather than a derived sampling distribution. What that makes of the posterior is an estimating equation, not a likelihood, and it is also why V is held fixed.

A study is not just its rows

Nothing in ϕ is indexed by cohort, so two labs reporting the same readout at different levels have nowhere to put the difference. Give each study one shift on its own location rows:

$$\hat{T}_B \mid \eta \sim \mathcal{N}\left(\tau_B(\phi) + \sum_s \eta_s \mathbf{1}_{B,s}^{\text{loc}}, V_B\right), \quad \eta_s \sim \mathcal{N}(0, \tau_\eta^2).$$

Gaussian and additive, so it marginalises exactly:

$$\hat{T}_B \sim \mathcal{N}\left(\tau_B(\phi), V_B + \tau_\eta^2 \sum_s \mathbf{1}_{B,s}^{\text{loc}} \mathbf{1}_{B,s}^{\text{loc}\top}\right).$$

- No coordinate added to ϕ , and the term is free of ϕ , so V stays frozen and no log det comes back.
- **One study supplies 20 of the 55 rows.** Its within-study contrasts keep full weight; only their common level is discounted.
- τ_η is fixed, not free: 22 studies against 55 rows, 13 of them reporting a single row a free shift would fit exactly, and 8 directions already spanned by Z .

Nobody printed an error bar

The observation equation needs V_B , and most sources give nothing to build one from. So predict it: resample the model's own cohort.

$$V_B = V_B^{\text{boot}} + E_B, \quad V_B^{\text{boot}} = \text{Cov}_* [T_B(\tilde{F}_B^*)]_{\phi_0}.$$

- Sampling noise from the predicted cloud, emulator error on top.
- Built once at a plug-in ϕ_0 , a fixed guess at the answer used only to size the error bars, and frozen there.

What drawing whole patients buys

- A resampled patient carries its whole vector $\tilde{x}_{ic}$. That is what moves the dependence between readouts into V_B , rather than leaving it to be asserted.
- Where cohorts of a block share people the draw is common, and that is the only source of across-cohort correlation.
- Two readouts of one cohort that share a denominator move almost in lockstep, which leaves V_B close to singular. A small ϵI is added before it is inverted.

The cost of all three: V_B now rests entirely on the predicted cloud.

The guard on that cost

A cloud that is too narrow makes every error bar too small, the likelihood then overrides the prior, and nothing in the fit catches it. So check first, against any source that does report a sampling uncertainty U_{rck} :

$$\varrho_{rck} = U_{rck} / \sqrt{(V_c^{\text{boot}})_{rck, rck}}.$$

- Above one, the predicted cloud is too narrow at that readout.
- Computed before fitting, so it is a prior predictive check on exactly the rows that now carry ω .
- **Fix the model rather than widening V_B :** widening hides what is being tested.

Emulator error

Draw L held-out clouds at ϕ_0 and evaluate each both ways:

$$d_B^{(\ell)} = \tau_B^{\hat{g}}(\phi_0; z^{(\ell)}) - \tau_B^g(\phi_0; z^{(\ell)}), \quad E_B = \text{Cov}_\ell [d_B^{(\ell)}].$$

- Defined on the *statistics*, not the readouts: the map from a readout error to a statistic error depends on the functional.
- A covariance only sees how the error *varies*. If the emulator is off by the same amount every time, E_B is blind to it, so **report the average error $\bar{d}_B$ alongside**. The second check later asks whether the discrepancy map can absorb it.

Where $\hat{g}$ is trained is part of the model

E_B and $\bar{d}_B$ are properties of $\hat{g}$, and $\hat{g}$ is a property of the values it was trained on.

- The fit calls it at $\mu + D_\omega L_R z_i$, so the values it will be asked about are spread with covariance about $\Sigma_1 + \Omega$. Train it on a design drawn from that same law, widened at the edges.
- Every parameter has to vary in the design. Pin one and the emulator has never seen it move, so the likelihood along it is not flat, it is undefined.
- Score it on a fresh draw, not on a held-out slice of the design. Space-filling designs are built so their halves interleave, so a split would test it on points it effectively already saw.

One number, one job

A reported uncertainty could either set the error bar on its row or say something about population width. It cannot do both.

- **It always goes to width.** No parameter here has a published between-patient spread, so width rows are the only thing that says anything about ω . Error bars, by contrast, the bootstrap gives us on every row for nothing.
- What we predict still depends on which number the paper printed. A standard deviation describes the patients; a standard error describes how well the study pinned down its own summary. **At $n_c = 10$ they differ by a factor of 3.2**, so the extraction has to record which one it is.
- Either way the row is worth having, and it does not make the ambiguity at the end of the deck any worse.

Priors and posterior

Priors: location and spread

$$\mu \sim \mathcal{N}_P(\mu_0, \Sigma_1)$$

from the stage-1 composite prior. Write $\mathcal{M}$ for the parameters whose between-patient spread some source has actually measured; today it is empty. For $j \in \mathcal{M}$, $\log \omega_j$ concentrates at $\log \omega_{0j}$ with precision set by that source's n_j . For the rest,

$$\log \omega_j = \log \omega_{0j} + s + u_j, \quad \sum_{j \notin \mathcal{M}} u_j = 0.$$

- s is how much wider the population is overall; u is the pattern across parameters.
- Without the constraint the two are not separately defined: add a constant to s , subtract it from every u_j , and every ω_j comes out identical. Summing the u_j to zero removes that freedom and leaves s as the only overall level.
- Worth the trouble: s is the part that reaches the virtual population, and the part the last slides are about.

Priors: discrepancy

Measurement. Free, because it has no parameter twin:

$$a \sim \mathcal{N}(0, \sigma_a^2 I), \quad b \sim \mathcal{N}(0, \sigma_b^2 I), \quad \sigma_a = \sigma_b = 0.5.$$

Mechanism. Tightly shrunk, because it competes with μ :

$$\beta_q = \tau_\beta (\beta_q^{\text{raw}} - \bar{\beta}^{\text{raw}}), \quad \tau_\beta = 0.15 \text{ fixed.}$$

for a declared handful of species, by default the shared denominators.

- Fixed rather than estimated: estimating τ_β gives the sampler a funnel-shaped posterior, which it handles badly, and with β centred there is too little left for it to learn in exchange.
- Centred because otherwise there is an exact alias. Give every species the same bias and it cancels out of every ratio and fraction, leaving only a shift in the densities, which γ already produces. The data could never tell those apart.

Declared assay conversions

Some readouts measure a different quantity from the species they are matched to, and say so: Herremans reports IL-10 in tumour homogenate, which is total, while $V_T\text{.IL10}$ is the free pool.

$$\log R_m \sim \mathcal{N}(\log R_{0m}, \sigma_{Rm}^2)$$

- They go inside h_r , the only place indexed by readout and species at once. β is shared across readouts and γ acts after the composition, so neither can express a conversion that applies to one readout only.
- The size is wrong for β anyway: $\tau_\beta = 0.15$ cannot carry a $\log R$ of 1.6 to 3.9, which is a factor of 5 to 50.
- Where a species has both a free β and a conversion, and no readout separates them, only one of the two may be free.

Posterior and deliverable

$$\phi = (\mu, s, u^{\text{raw}}, \log \omega_{\mathcal{M}}, a, b, \beta_S^{\text{raw}}, \log R_{\mathcal{A}}), \quad p(\phi \mid \mathcal{D}) \propto p(\phi) \prod_{B \in \mathcal{B}} \mathcal{N}(\hat{T}_B; \tau_B(\phi), V_B).$$

Reading ϕ left to right: where the population sits, how wide it is overall, the per-parameter pattern of widths, any parameter with a measured spread, the assay location and scale terms, the species biases, and the declared conversions.

The virtual population is the posterior predictive over patients:

$$p(\theta_{\text{new}} \mid \mathcal{D}) = \int \mathcal{N}_P(\log \theta_{\text{new}}; \mu, \Omega) p(\phi \mid \mathcal{D}) d\phi,$$

emitted as an **ensemble**, one population per posterior draw, so that between-patient variability and uncertainty about ϕ stay separable downstream.

Fitting it, and checking it

Why fit a flat model first

Everything except the shape of the population is already in play: the forward map, the discrepancy terms, the block structure, the sampler.

- Hold ω at ω_0 and all of that can be tested cheaply, on the centre rows alone.
- Every width row then becomes genuinely held out.
- And it hands back the plug-in ϕ_0 that V_B needs anyway.

A fault the flat fit turns up is one the population fit would have had too, found for a fraction of the cost.

The flat fit

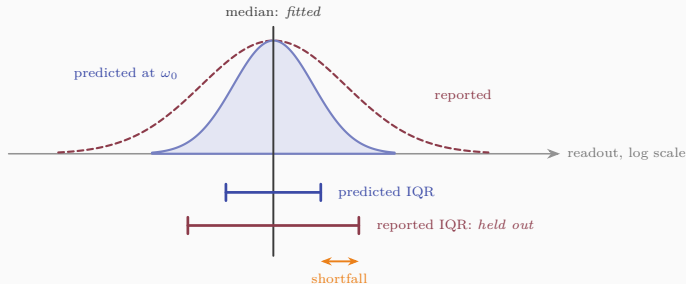
$$\phi_{\text{flat}} = (\mu, \mathbf{a}, \beta_{\mathcal{S}}^{\text{raw}}), \quad \omega = \omega_0, \quad b = 0.$$

Nothing else changes. It is optimised rather than sampled, and the uncertainty in $\hat{\mu}$ comes from the curvature at the optimum, which is all this fit has to deliver.

Why ω_0 and not zero. Collapsing the cloud onto a point kills three things at once:

- the bootstrap returns zero, so V_B has to come from outside;
- eligibility becomes a switch on the whole cohort, discontinuous in μ ;
- every location functional returns the same number, so a mean and a median predict one value for two rows.

The width shortfall



The flat fit estimates μ at $\omega = \omega_0$ from the centre rows alone, so every width row is a genuine out-of-sample prediction.

A reported quartile is a width row too. Fitting $\hat{q}_{0.25}$ and $\hat{q}_{0.75}$ would leave the comparison measuring a residual from a fit that had already seen it.

What the flat fit buys, and what it cannot test

V_B is computable	The bootstrap resamples a real cloud.
Width rows are held out	Predict them at ω_0 and report the ratio per readout.
$\hat{\mu}$ needs no conversion	It is already the centre of a population, so nothing has to be adjusted afterwards.
It supplies the plug-in	$\phi_0 = (\hat{\mu}_{\text{flat}}, \omega_0)$.

It cannot test the shape of the population: ω , R , and the direction s and b_1 share.

Most of ϕ is not identified. Which parts moved?

ϕ has far more coordinates than there are rows, so most of the posterior is the stage-1 prior handed straight back. That is the design rather than a failure. What is worth knowing is which directions the rows actually moved.

$$G = I + J^\top J, \quad J = V^{-1/2} \frac{\partial \tau}{\partial \phi} \Sigma_\phi^{1/2}.$$

J is how much each predicted row moves when each parameter moves, put on a common scale: divided by that row's error bar, multiplied by that parameter's prior width. G is then the posterior precision, and the I is what the prior on its own would give.

- J has only K rows, so at least $\dim \phi - K$ directions have $Jv = 0$. Move along one of those and not a single prediction changes, so the posterior there is exactly the prior.
- Report the eigenvalues of G , and how $\text{tr}(J^\top J)$ splits across the blocks of ϕ .

Two consequences of the shape alone

- **The measurement map takes most of the information.** a has a handful of coefficients against hundreds of mechanism parameters, and acts directly on the rows where the mechanism must act through the simulator. So whatever the model can route into a will end up there, the emulator's offset included.
- **Row space is saturated.** When $\text{rank}(J)$ equals the number of rows, “can the model produce this row effect” is answered yes by construction. A useful check therefore has to ask what an effect costs, not whether it is reachable.

Check: can the map hold the emulator's offset?

To first order a shifts the location rows and b the scale rows. Stack those effects into W : that is everything the measurement map is able to do to the data. Now ask how much of the emulator's average error $\bar{d}$ lies inside that span, measured in error-bar units:

$$\text{captured} = \|P_{V^{-1/2}W} V^{-1/2}\bar{d}\| / \|V^{-1/2}\bar{d}\|.$$

A high fraction says the emulator's bias is representable, and is being paid for in a . A low one says it is sitting in μ instead.

Check: does Z have work of its own?

The model claims assay modality and quantity type have no mechanistic twin. The obvious test, whether the mechanism could reproduce a column of Z , says yes every time: with hundreds of parameters against tens of rows the mechanism can hit any row pattern at all. So ask what it would *cost* instead, as the smallest mechanism move that imitates that column:

$$\delta_j^* = \arg \min_{\delta} \delta^\top \Sigma^{-1} \delta \quad \text{subject to} \quad J_{\mu\omega} \delta = W e_j.$$

Report $\|\delta_j^*\|$ in prior standard deviations. Large means only an implausible mechanism move imitates that column, so the column is identified. Small means the two are confounded, and the priors decide the split.

Where I am least sure

What the fixed cloud has that we do not

Weight collapse is at least a joint signal. One weight vector has to satisfy every observable at once, so what strain it shows reflects the *coupling* between observables. A parametric fit always returns something, and has no equivalent of its own.

A hybrid. Reweight the fitted $F(\theta \mid \hat{\phi})$ rather than a wide prior. Starting from a population the data already agree with removes most of the collapse that dimension alone causes, so the strain that remains is attributable, and **measures how much shape the lognormal missed.**

Known simplifications

- **Independence across blocks.** Shared patients are handled inside a block, so what remains is that no two publications overlap, which registry sources do not guarantee.
- **The study offset is shrunk, not free.** τ_η is asserted, and only 7 of the 17 (quantity type, assay modality) cells are reported by more than one study, so the corpus constrains it weakly. Report the fit at more than one τ_η .
- **V frozen at ϕ_0 .** Letting it move with ϕ brings back a $-\frac{1}{2} \log \det V$ term that rewards a narrow population whether or not it fits. The cost is that the error bars do not track ϕ .
- **Z is asserted, not tested.** The cost computation above is the version that survives the dimensions, and it has not been run.

R is an input, and it reaches ω

The factor form makes R cheap to state and positive definite by construction, but nothing pins the loadings and R carries no uncertainty. Joint and tail questions rest on it entirely.

It reaches the marginals too. Correlated parameters pushing a readout the same way inflate its variance, so a diagonal R under-predicts spread and the fit inflates $\hat{\omega}$ to meet the published IQRs.

Fit at both. If $\hat{\omega}$ barely moves, the marginal widths are safe and R touches only joint statements. If it moves, they are partly an artefact.

The one that changes the answer

When the fitted population comes out narrow there are two explanations, and the data do not separate them:

- the patients really are alike (s), or
- the simulator under-disperses and κ has stretched the readouts to compensate (b_1).

Only s reaches the virtual population. b_1 is discarded with the rest of the measurement map, so the split decides the answer.

s against b_1 : what would break the tie

- A width row responds to b_1 the same way whatever it measures. It responds to s only through the parameters whose spread is not already known. So a readout separates the two only if its balance between those differs from the other readouts'.
- Right now no parameter has a measured spread at all, so that balance is identical everywhere and nothing separates them. The priors τ_s and σ_b decide the answer between them. Report the joint posterior of the pair so the split is at least visible.
- A prior on assay measurement error would break the tie, and an assay CV is far easier to come by than a measured between-patient width.

Backup: indices and dimensions

	Range	
j	$1, \dots, P$	model parameters
q	$1, \dots, Q$	model species
r	$1, \dots, M$	readouts
s	$1, \dots, S$	scenarios
c	$1, \dots, C$	study cohorts; $s(c)$ is the scenario cohort c was run under
k	$1, \dots, K_{rc}$	statistics that cohort c reports for readout r
i	$1, \dots, N$	simulated patients
$\mathcal{R}_c$	$\subseteq \{1, \dots, M\}$	the readouts that cohort c reports
K_c	$\sum_{r \in \mathcal{R}_c} K_{rc}$	the rows that cohort c contributes
B	$\in \mathcal{B}$	a block: the cohorts that have to be drawn together
K_B	$\sum_{c \in B} K_c$	the rows of block B

Backup: fixed inputs

	value	
ω_0	—	Spread centre, per parameter. Elicited by role.
Λ, ψ	—	Loadings, bins 0.8/0.5/0.3 with a sign. Communality capped at 0.81.
R	—	Population correlation, $\Lambda\Lambda^\top + \text{diag}(\psi^2)$. Not identified by these data.
μ_0, Σ_1	—	Stage-1 composite prior, log scale.
h_r	—	Readout composition, including unit conversions.
Z	—	Readout attributes: assay modality, quantity type.
c_{rc}	—	Pivot: cohort c 's median of readout r at (μ_0, ω_0) .
S	—	Species allowed a mechanism discrepancy. Default: the shared denominators.
$z_{1:N}$	—	The latent cloud, drawn once.
$w^{(c)}$	—	Eligibility weights, read from each paper's criterion.
h_q, h_w	—	Smoothing bandwidths: the quantile kernel and the eligibility weight.
$\hat{g}$	—	The emulator, and the design it was trained on. Fixes E_B and $\bar{d}_B$.
V_B	—	Assembled at ϕ_0 , then frozen. One per block.
E_B	—	Emulator error covariance on the statistics of block B .
τ_s	0.30	Global level of the spreads.
τ_ω	0.085	Pattern across spreads. Derived: a quarter of the role ladder's narrowest gap.
τ	0.50	Trust in the role prior against a measured spread.
σ_a, σ_b	0.50	Measurement discrepancy per column of Z .
τ_β	0.15	Mechanism discrepancy scale, fixed rather than estimated.
τ_η	—	Study effect scale. Fixed rather than estimated; not yet set.

Backup: the finite- n gap, in full

At a log-scale spread of 1 and $n_c = 9$:

population $q_{0.25}$	-0.674	what the statistic estimates
$\mathbb{E}[\hat{q}_{0.25}]$	-0.570	what a paper reporting nine patients prints
spread matching	-0.570 0.845	what the fit returns if it predicts the population functional

The gap runs inward on both sides, so a quartile pair pulls Ω down from both directions. It closes as n_c grows, not as the predicted cloud does.